Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Ionomer-Free Porous Transport Electrodes for Next-Generation Anion Exchange Membrane Water Electrolyzers*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Mechanical · Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*Surface Treatment*	*Coating/ Method Technologies*
Concept of Proposed Technology	The proposed technology is to enhance the durability of anion exchange membrane water electrolyzers (AEMWEs) through the application of an ionomer-free porous transport electrode (PTE) to set AEMWEs as the mainstream technology for clean hydrogen production. The durability of AEMWEs remain severely limited compared to competing electrolyzer technologies [1], largely owing to the ionomeric binders. Although binders are essential for inter-catalyst particle contact and hydroxide ion transport, the ionomers undergo oxidation during AEMWE operation and cause a reduction in electrochemically active sites [2]. While significant efforts have been geared towards developing alternative ionomeric binders [3], an ionomer-free electrode is a promising approach to not only mitigate ionomer-related durability issues but also to simplify the manufacturing process of			

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	electrodes [4]. Despite their promise, significant challenges related to ionomer-free PTEs remain, including low electrochemical surface area [5], undeciphered mass and ion transport mechanism [4], and durability of electrode materials [6], [7], such as dissolution and redeposition. We propose a holistic approach of designing an ionomer-free PTE, where the PTE design will be guided by numerical modelling and X-ray-based characterization techniques to ensure optimal transport and contact properties of the PTEs. The meticulously tailored ionomer-free PTE will undergo comprehensive electrochemical testing for performance and durability to evaluate its feasibility at industrial relevant conditions.
Rationale for Joint R&D Project	Neel Sciences is an emerging SME in the field of green hydrogen, with the capabilities of scaling up production of PTEs to industrial-scale applications. The proposed joint project between Toronto Metropolitan University (TMU), University of Victoria (UVic), and Neel Sciences will enable the world's first ionomer-free PTE-based AEMWE testing in an industrial scale (active area > 25 cm^2). Professor Jason K. Lee from UVic is a world-renowned expert in numerical simulations of transport phenomena within PTLs [8], [9] and design and fabrication of ionomer-free porous transport electrodes via physical vapor deposition methods [10]. Professor ChungHyuk Lee from TMU has significant experience in micro X-ray fluorescence imaging and X-ray radiographic techniques to characterize PTL/electrodes for fuel cells and electrolyzer applications [11], [12], [13] and designing patterned electrodes for enhanced performance and durability [14]. Combining the team's expertise in characterization, design, and testing of PTEs combined with Neel Sciences's expertise in scaled-up production of electrodes and testing of electrolyzers, we will be able to accomplish a new paradigm in PTEs for AEMWEs at an industrially relevant scale.
Key Words	1. Porous transport electrodes without ionomeric binders for enhanced durability of AEMWEs 2. X-ray-based characterization techniques for understanding the homogeneity of catalyst coating 3. A three-dimensional modelling platform for elucidating key properties of PTEs for AEMWE applications

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	The objective of this project is to demonstrate a scalable ionomer-free PTE with enhanced performance and durability for AEMWE applications.			
R&D Target Specifications	To ensure the feasibility of the ionomer-free PTE, industrially relevant performance and stability targets must be closely achieved [15]. The developed ionomer-free PTE will demonstrate electrochemical performance of 1.0 A/cm² at 1.8 V and exhibit stability of < 1 mV/h after 500 h of operation at 0.6 A/cm².			
R&D Plan	We will develop ionomer-free PTEs using two approaches: (i) an electrodeposition approach, and (ii) a physical vapor deposition (PVD) approach. Specifically, for the electrodeposition approach, we will utilize a high-throughput optimization platform where optimal electrodeposition conditions (electrolyte concentration, electrical potential, and temperature) will be identified experimentally. Additionally, for the PVD approach, we will employ sputtering techniques available at UVic. Simultaneously, we will develop an integrated AEMWE testbed platform for performance and durability evaluation. We will employ X-ray-based characterization techniques (i.e., micro X-ray fluorescence imaging and X-ray micro-computed tomography) to elucidate the physicochemical properties of the developed PTEs, such as the pore size distribution, catalyst coating distribution. We will also utilize stochastic - pore network coupled modelling to create novel PTE structures and characterize the two-phase transport properties that occur during operation. Moreover, we will utilize 3D finite-element-analysis modelling to quantify the mechanistic properties (i.e., tensile/compressive stress distribution, contact area, protrusion/indentation) under ionomer-free configuration. Based on the finding of the models and X-ray characterization, we will inform the design of the ionomer-free PTEs and narrow down our PTE fabrication technique to the more appropriate technique. We will also perform electrochemical testing (i.e., polarization analysis, electrochemical impedance spectroscopy, accelerated stress tests) using a single-cell AEMWE testing apparatus. The developed ionomer-free PTEs will be evaluated and tested by Neel Sciences on industrially relevant conditions. We will accomplish our R&D objectives through the proposed milestones (M) over the next 3 years: **M1**: Demonstrate feasibility of ionomer-free PTE fabrication using sputtering an electrodeposition techniques, and design of a 25 cm² active-area AEMWE stack and selection of key stack components compatible with ionomer-free porous transport electrodes (*Year 0.5*) **M2-1**: Go/No-Go decision on sputtering-based or electrodeposition-based approach of fabricating ionomer-free PTEs (or a combination of two techniques) (*Year 1.5*) **M2-2:** Fabrication of a 25 cm² active-area AEMWE stack and optimization of electrochemical performance under laboratory-scale operating conditions (*Year 1.5*) **M3:** Establishment of an integrated AEMWE stack-level test-bed environment for performance and			

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	durability evaluation under controlled operating conditions (*Year 2.0*) **M4-1**: A pore network model-based computational platform for simulating the effects of PTE structure and tensile/stress distribution effects on the performance of AEMWEs with ionomer-free PTEs (*Year 2.5*) **M4-2**: An X-ray based characterization platform for investigating the catalyst coating uniformity and internal pore structure of 25 cm²-active area PTE (*Year 2.5*) **M4-3:** Technology transfer of ionomer-free PTE fabrication processes from global partner research institutes and evaluation of stack-level performance with integrated PTEs (*Year 2.5*) **M-5**: Long-term durability validation of AEMWE stacks incorporating ionomer-free PTEs through global collaborative testing, and development of commercialization strategies for industrial deployment (*Year 3.0*)

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The roles of TMU are (1) Establish an electrodeposition-based PTE fabrication platform for AEMWE applications, (2) Perform X-ray micro-computed tomography and X-ray fluorescence imaging to characterize the physicochemical properties of the PTEs, (3) Perform electrochemical testing of AEMWEs on a single-cell test bench. The roles of UVic are (1) Establish a sputtering-based PTE fabrication platform for AEMWE applications, (2) Construct a stochastic - pore network coupled modelling tool for studying the two-phase transport properties within the PTEs, and (3) Develop a 3D finite-element-analysis modelling tool to quantify the mechanistic properties of the PTEs.
Expected R&R of Korean SME Partner	The role of Neel Sciences is to serve as the industrial implementation and validation partner in the proposed international joint R&D project. The specific roles and responsibilities of Neel Sciences are as follows: (1) Securing and optimizing ionomer-free PTE fabrication technologies for AEMWE applications developed by global partner research institutes. (2) Establishing a testing and evaluation system capable of objectively assessing performance and durability under industrially relevant operating conditions. (3) Integrating the developed ionomer-free PTEs into AEMWE stacks and conducting long-term operational testing to evaluate stack-level performance and durability. (4) Promoting the commercial productization of AEMWE stacks that incorporate outcomes from the global collaborative research and formulating strategies for industrial deployment and market expansion. Through these roles, Neel Sciences will function as a critical bridge between fundamental research outcomes and industrial application, enabling effective technology transfer and successful commercialization.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	To date, research on AEMWE technology in Korea has primarily focused on the development of non-precious metal catalysts and the enhancement of membrane durability through reinforced composite membranes. However, performance degradation arising from ionomer oxidation during electrode fabrication, which leads to the detrimental loss of electrochemically active sites, has not been sufficiently addressed. The technology proposed in this project aims to fundamentally resolve this critical limitation through the development of ionomer-free PTEs. By eliminating ionomer-related degradation mechanisms at the electrode level, this approach introduces a new pathway for significantly improving the durability of AEMWE systems. Furthermore, by integrating advanced electrode design strategies, X-ray-based diagnostic and characterization techniques, and stack-level validation under industrially relevant operating conditions, this project is expected to establish a new technological paradigm for durable and scalable AEMWE systems. From a national perspective, the successful execution of this project will strengthen Korea's technological competitiveness in next-generation hydrogen production technologies and accelerate the transition of AEMWE systems from laboratory-scale research to industrial implementation. In addition, through global collaborative R&D, the project will enhance the technological self-reliance of Korean SMEs, contribute to the creation of high-value intellectual property, and support future standardization efforts. Ultimately, the outcomes of this project will provide a robust foundation for the commercialization of competitive AEMWE systems in both domestic and global markets.

6. References

[1] D. Li *et al.*, "Durability of anion exchange membrane water electrolyzers," *Energy Environ Sci*, vol. 14, no. 6, pp. 3393–3419, 2021.

[2] E. K. Volk *et al.*, "Role of the Ionomer in Supporting Electrolyte-Fed Anion Exchange Membrane Water Electrolyzers," *ACS electrochemistry*, vol. 1, no. 2, pp. 239–248, 2024.

[3] J.-H. Choi *et al.*, "Performance-Limiting Factors of Hydrocarbon Ionomeric Binders for Fuel Cells and Electrolyzers," *ACS Energy Lett*, vol. 10, no. 5, pp. 2392–2399, 2025.

[4] M. Chen, Y. Ren, C. Lee, and J. K. Lee, "Ionomer-free electrodes in PEM water electrolyzers: a critical review," *Journal of Physics: Energy*, 2025.

[5] P. Kúš *et al.*, "Magnetron sputtered thin-film vertically segmented Pt-Ir catalyst supported on TiC for anode side of proton exchange membrane unitized regenerative fuel cells," *Int J Hydrogen Energy*, vol. 44, no. 31, pp. 16087–16098, 2019.

[6] J. K. Lee *et al.*, "Nanochannel electrodes facilitating interfacial transport for PEM water electrolysis," *Joule*, vol. 8, no. 8, pp. 2357–2373, 2024.

[7] M. Yasutake *et al.*, "Ru-core Ir-shell electrocatalysts deposited on a surface-modified Ti-based porous transport layer for polymer electrolyte membrane water electrolysis," *Int J Hydrogen Energy*, vol. 49, pp. 169–183, 2024, doi: https://doi.org/10.1016/j.ijhydene.2023.07.048.

[8] J. K. Lee, C. H. Lee, and A. Bazylak, "Pore network modelling to enhance liquid water transport through porous transport layers for polymer electrolyte membrane electrolyzers," *J Power Sources*, vol. 437, p. 226910, 2019, doi: https://doi.org/10.1016/j.jpowsour.2019.226910.

[9] J. K. Lee and A. Bazylak, "Optimizing porous transport layer design parameters via stochastic pore network modelling: reactant transport and interfacial contact considerations," *J Electrochem Soc*, vol. 167, no. 1, p. 013541, 2020.

[10] J. K. Lee *et al.*, "Ionomer-free and recyclable porous-transport electrode for high-performing proton-exchange-membrane water electrolysis," *Nat Commun*, vol. 14, no. 1, p. 4592, 2023.

[11] C. H. Lee *et al.*, "Temperature-dependent gas accumulation in polymer electrolyte membrane electrolyzer porous

transport layers," *J Power Sources*, vol. 446, p. 227312, 2020, doi: https://doi.org/10.1016/j.jpowsour.2019.227312.

[12] C. Lee *et al.*, "Toward a comprehensive understanding of cation effects in proton exchange membrane fuel cells," *ACS Appl Mater Interfaces*, vol. 14, no. 31, pp. 35555–35568, 2022.

[13] A. P. C. Lee *et al.*, "Elucidating Cation Transport Properties in Nafion Membranes and Electrode Ionomer Network via X-ray Fluorescence Imaging," *J Electrochem Soc*, vol. 172, no. 6, p. 064502, 2025.

[14] C. Lee *et al.*, "Grooved electrodes for high-power-density fuel cells," *Nat Energy*, vol. 8, no. 7, pp. 685–694, 2023.

[15] Y. J. Son, G. Mirshekari, S. J. Raaijman, and P. J. Corbett, "Technology Landscape of Anion Exchange Membrane Water Electrolyzers: Where Are We Today?," *ACS Energy Lett*, vol. 10, pp. 3058–3063, 2025.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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